

Magnet / Hall sensor engineering package

Excel calculation tool, printable calibration fixture, and electrical test drawings.

Start with **Magnet_Hall_Calculator.xlsx**. The yellow cells contain clearly labeled example measurements. Replace them with your two magnet readings, standard-magnet data, actual dimensions, sensor approach/release readings and uncertainty bounds. The workbook is macro-free and requires no Solver add-in.

Workbook area	Purpose
Inputs + Calibration	Enter magnet dimensions and delivery grade, isolated surface gauss, reference magnet details, operating temperature, installed travel and measured sensor ON/OFF gaps.
Results + Distance curves	Read nominal switch distances, bounded ON interval, OFF distance, remaining gap allowance and the endpoint/midpoint state checks.
B-H model	Native editable recoil curves for both magnets with marked conditional switch-loss points. Effective permeance coefficients are user assumptions.
Aging	Optional logarithmic common-retention scenario. Without aging measurements, time to failure remains undetermined.

Geometry assumed

The analytic model assumes the magnet axes are aligned with horizontal travel and each sensor faces an outward south pole. The two inward north faces are separated by a nonmagnetic spacer. The remote magnet subtracts from the field at the tested sensor. This is a fixed-magnetization superposition model; mutual demagnetization needs an assembled field scan or magnetic analysis.

The user confirmed the state sequence: maximum = max ON / min OFF; midpoint = both OFF; minimum = max OFF / min ON. Magnetic ON is a LOW electrical output. The model suppresses axial conclusions if Inputs is set to Lateral or Unknown.

What the files establish

The design has been checked digitally; it has not been printed or bench-qualified. The workbook passed 11 boundary/reference tests, its final formula scan and its chart-link checks. Individual printed-part meshes are watertight and STEP solids are valid. Clearance checks cover three carriage positions. These checks do not establish as-built accuracy or magnet life.

Lifetime and B-H interpretation

Initial surface gauss, N35 grade and temperature do not determine aging rate or the true demagnetization knee. The B-H curves are linear effective-load approximations. A time estimate appears only after entering a qualified retention measurement and test duration; both magnets must follow the same normalized loss law. Extrapolation is identified. Unequal losses, corrosion and irreversible thermal damage require additional data.

Build the calibration fixture

Mechanical purchase and print list

Ref / qty	Part	Make or buy
P01 / 1	Base	3D print rigid nonmagnetic polymer
P02 / 1	Block1 carriage	3D print rigid nonmagnetic polymer
P03 / 2	Carriage keepers	3D print rigid nonmagnetic polymer
P04 / 1	Block2 sensor mount	3D print rigid nonmagnetic polymer
P05 / 1	3 mm inner-face spacer	3D print; 1 and 6 mm alternatives included
P06 / 2	Magnet retaining cups	3D print; 6.65 mm pocket / 5.80 mm aperture
P07 / 1	Return band (optional)	3D print TPU 95A, 0.8 mm thick
P08 / 1	Nominal 5 mm datum gauge	3D print; calibrate actual contact separation
P09 / 4	Table feet	3D print; clearance for screw ends
H01 / 1	0-13 mm micrometer head	McMaster 19725A12 / Mitutoyo 148-504
H02 / 2	M3 x 0.5 x 10 mm nylon screws	McMaster 96295A708; cassette foot
H03 / 6	M3 x 0.5 x 12 mm nylon screws	McMaster 96295A709; sensor/support/cups
H04 / 4	M3 x 0.5 x 20 mm nylon screws	McMaster 96295A712; keepers
H05 / 12	M3 x 0.5 nylon hex nuts	McMaster 93800A400; AF 5.4 x 2.2 mm

The supplied sensor and magnets are the specified articles under test. All added fixture components are printed or McMaster COTS. Screw packages contain more than the installed quantities. For the 6 mm spacer, use two additional 96295A712 cup screws. Micrometer clamp hardware is supplied with the head. The CAD fastener and instrument envelopes are simplified.

Print guidance

Use a stable rigid nonmagnetic polymer such as PETG for the room-temperature prototype. Use at least four perimeters, 40% infill, and 0.10 mm layers on the cups. Orient cups with the retaining face on the bed and the pocket upward; orient the base and carriage flat. Use supports under horizontal holes/bridges where needed and remove them without changing datums. The STL files retain drawing coordinates; reorient and place on the bed in the slicer. Do not print the assembly-reference STL.

Check each fit with the actual parts before completing the build. The specified disk tolerance is ± 0.1016 mm, while the cup is nominal. Adjust cup depth to measured magnet thickness, and measure actual seated spacing. Do not force a magnet into a tight pocket. Printed ± 0.20 mm is a prototype target, not a calibration accuracy.

Assembly and adjustment

1. Prepare base and carriage

Clean the 30.60 mm channel and 30.00 mm carriage sliding surfaces. Insert the nylon nuts into the base and carriage underside pockets. Install four table feet. Slide the carriage onto the base and install the two keepers with four M3 x20 screws. The second keeper is rotated 180 degrees. Tighten only enough to retain the carriage while allowing smooth motion.

2. Install the micrometer

Secure the 9.525 mm micrometer hub in the printed 9.70 mm bore with its supplied clamp nut. The nominal bore axis is $Y=0$, $Z=30$ mm. Bring the spindle against the carriage push face using light consistent force. The long carriage keeps the steel micrometer approximately 90 mm or more from the magnets in the supplied pose; verify its field effect by comparison.

3. Make the removable pair cassette

Choose the spacer matching the measured inward-face separation; the default is 3 mm. Seat each disk in a retaining cup with its north face toward the spacer and south face outward. Fasten the cup/spacer sandwich using two M3 x12 screws and nuts. Attach the cassette foot to the carriage using two M3 x10 screws and captive nuts. The opposed pair repels, so control both cups while closing the stack.

4. Mount the sensor

Attach the 55100 housing to block2 with two M3 x12 screws and nuts, then attach block2 to the base with two more M3 x12 screws. Use the datum holes shown on P04. Cable exit is +Y; the sensitivity center is offset 3.5 mm from housing center so that it lies on the magnet axis. Perform a pole and lateral-alignment check before recording data.

5. Preload and establish the gap

The optional printed TPU band stretches between the two rear posts and applies light return force. If it cannot maintain smooth contact, use light consistent manual preload and include the resulting repeatability. Measure the actual 5 mm gauge thickness and use it to establish a known magnet-face-to-case gap, then remove it. The cup lip is 0.40 mm ahead of the magnet face; measuring to that lip introduces an offset.

6. Step and record

The 13 mm head stroke supports approximately 0.5-13.4 mm usable face gap with the supplied geometry. Approach/retract in 0.25 mm steps to locate a transition and then in 0.01 mm increments only if the verified uncertainty supports that resolution. Never drive the cup into the sensor. The alternative base holes shift block2 by -12 mm for a larger gap range.

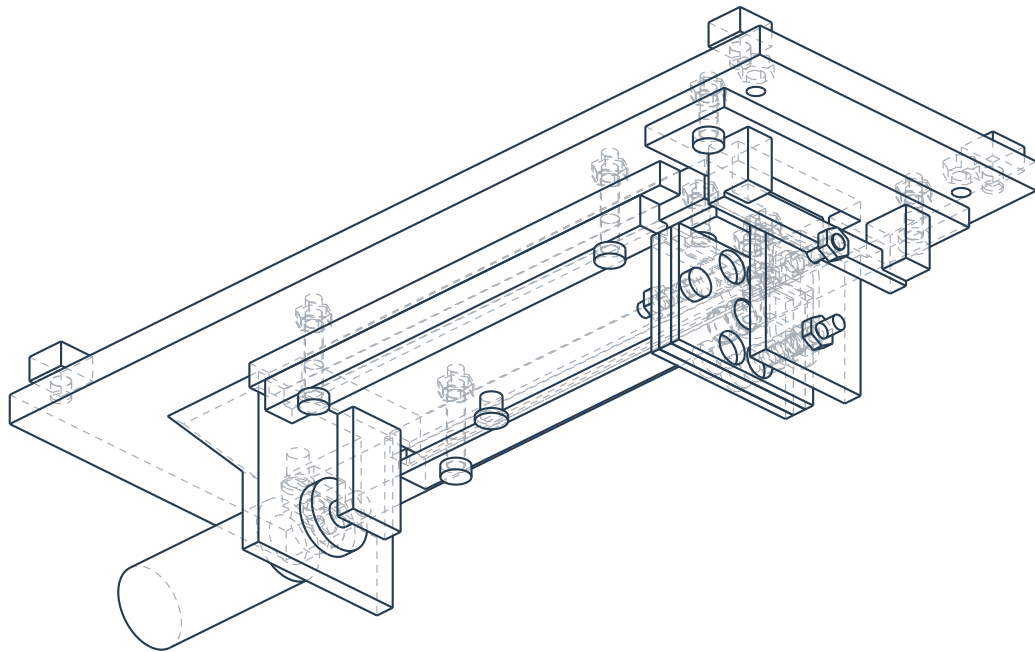
7. Test the second side or the standard

Rotate the entire cassette 180 degrees about vertical Z to present the other outward south face. Re-zero after every cassette/sensor change. For single-standard calibration, install one standard disk and leave the remote cup empty. Label the physical magnets and sensor serials; CAD near/far labels do not assign your production max/min sensor identities.

The printed spacer matches the static permeability role of aluminum; it does not reproduce aluminum thermal expansion, heat conduction, bond behavior or dynamic eddy currents. Use a qualified high-temperature fixture for thermal testing; this printable bench design is not rated to the magnet or sensor temperature limits.

A01 | AXIAL TEST FIXTURE - ASSEMBLY

Prototype Rev A | Units mm | Dimensions control; do not scale printed pages



ISOMETRIC / hardware envelopes simplified (view scale 0.74:1)

- 1 P01 base with remote micrometer panel
- 2 P02 block1 carriage + removable pair cassette
- 3 P03 keepers, two identical prints
- 4 P04 block2, sensor at offset sensitive center
- 5 P05 spacer; 3mm nominal supplied
- 6 P06 disk retaining cups, two prints
- 7 McMaster19725A12 micrometer head
- 8 P07 optional TPU return band (not shown)

Illustrated magnet face gap $g = 6.9125\text{ mm}$

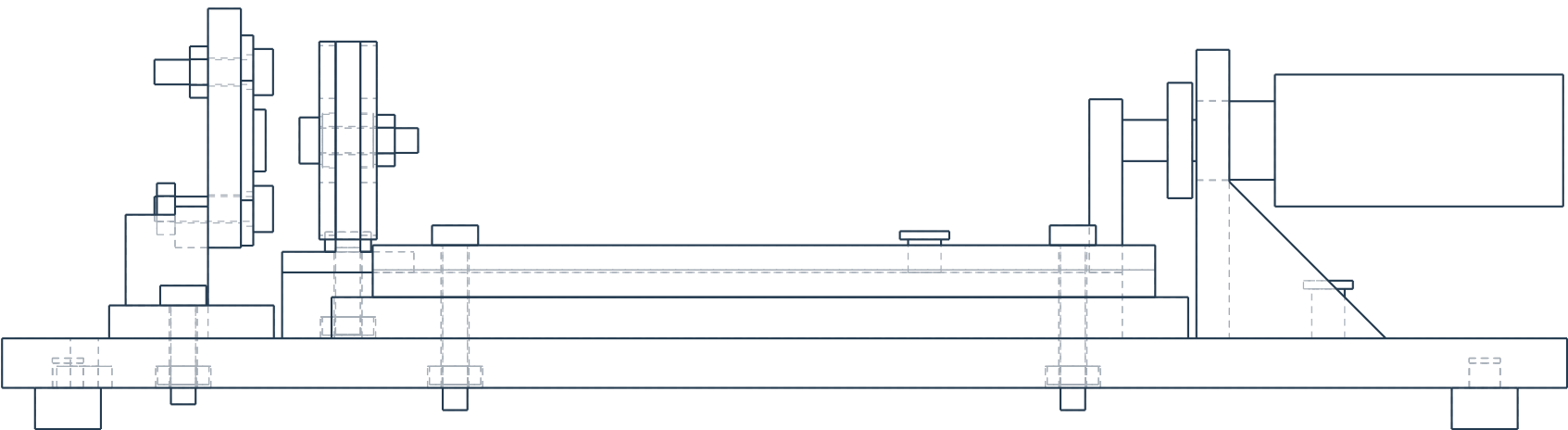
Fine stroke 13mm; typical usable $g \geq 0.5\text{ mm}$

Alternate block2 location shifts sensor -12mm

Observe south pole toward active sensor face

0.01mm head division is NOT fixture accuracy

Re-zero and measure actual face gap every setup



SIDE / X travel toward sensor (view scale 1.18:1)

190 BASE

A02 | PAIR CASSETTE, SENSOR DATUMS AND GAP DEFINITION

Prototype Rev A | Units mm | Dimensions control; do not scale printed pages

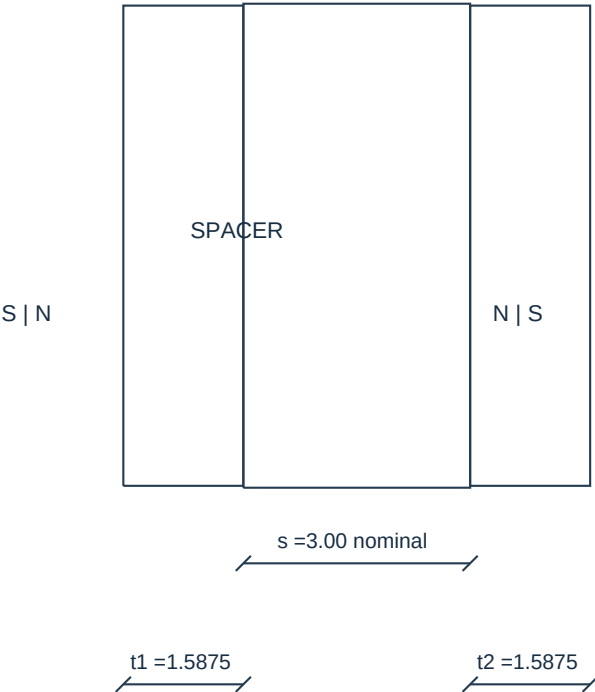
AXIAL SECTION - opposite magnetizations; both exterior faces SOUTH.

Use the actual seated inner-face separation s in the workbook, including adhesive/clearances.

Retaining cup lip is0.40 ahead of the exposed magnet face; central opening is DIA5.80.

The face gap g runs from exposed magnet surface to the selected sensor housing face.

Hall die depth inside the sensor is separate and must be calibrated or manufacturer-confirmed.



Sensor body:25.50 long x11.00 wide; thickness3.00 maximum.

Sensitive center is nominal2.00 from the edge opposite cable exit.

It is centered longitudinally:12.75 from either end; holes16.60 apart.

The drawing dimension7.50 is central raised-pad length, not die position.

Nominal sensor faceX32 and magnetic axisY0,Z30 are fixture datums.

Set the actual active face by a south-pole functional check.

Rotate the complete cassette180deg around Z to test magnet2.

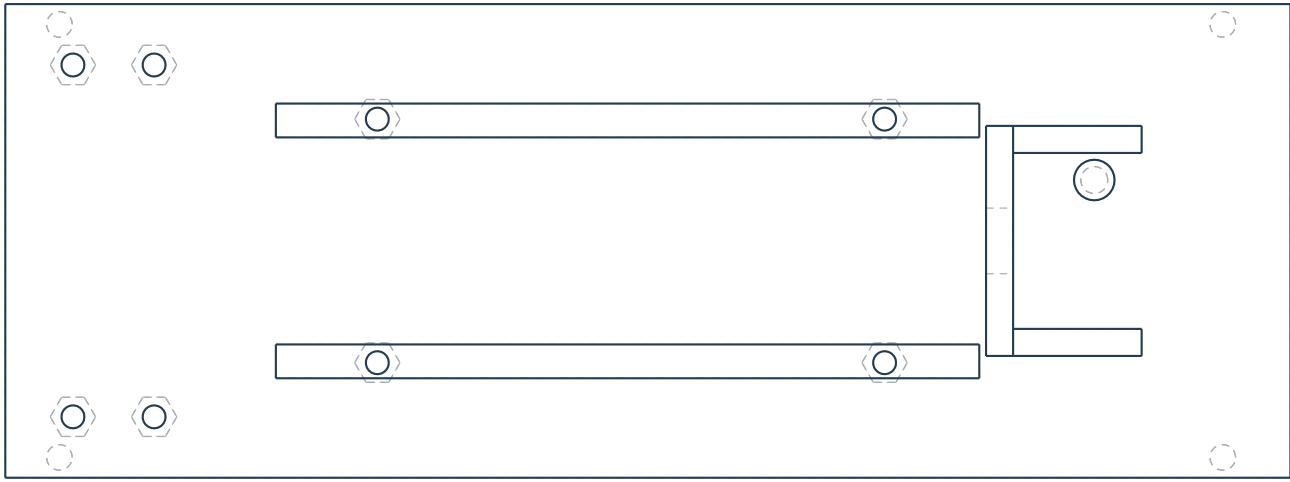
Re-establish actual g after rotating, changing spacer, or swapping sensor.

Calibration must include the remote opposing magnet when in pair mode.

For standard-single-magnet runs, leave the far cup empty.

P01 | P01 BASE

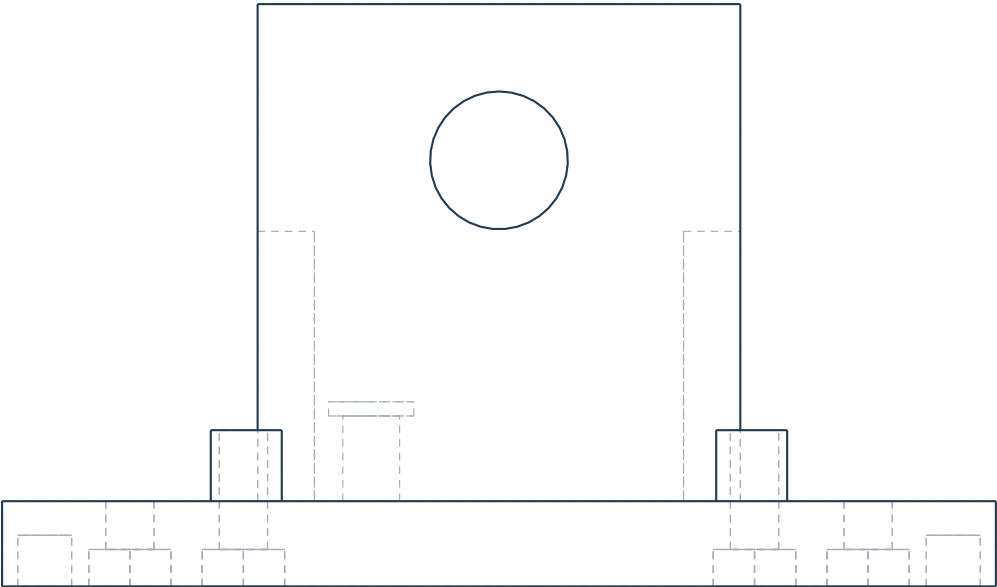
Prototype Rev A | Units mm | Dimensions control; do not scale printed pages



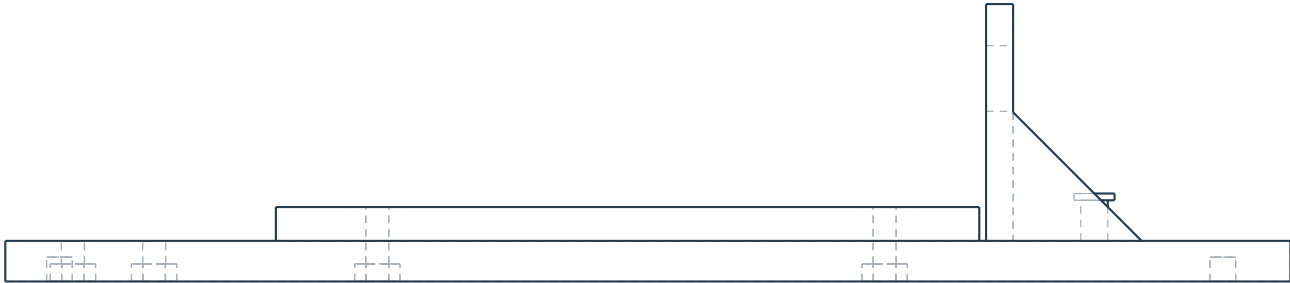
TOP (view scale 0.89:1)

190.0000 X

70.00 Y



END (view scale 1.88:1)



FRONT (view scale 0.89:1)

41.00 Z

BASE 190 x 70 x 6; print Z up, base flat.

Channel: 30.60 inside; rails 5 high, X40 to144.

Micrometer bore DIA9.70 at Y0 Z30; panel X145-149.

8x DIA3.40 through; underside nut recess AF5.80 x2.60.

Sensor X22 or10,Y+/-26; keepers X55,130,Y+/-18.

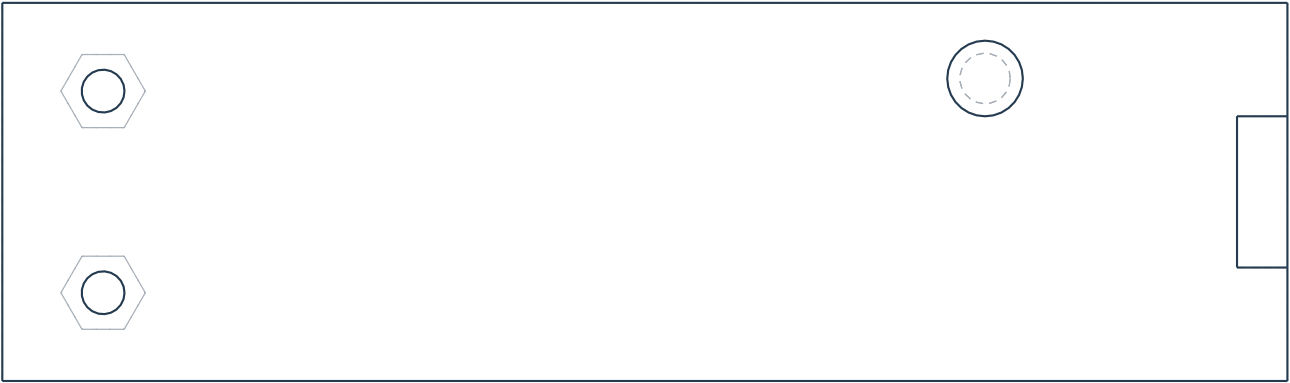
Feet: DIA3.80 blind3.60 at X8,180; Y+/-32.

General prototype target: +/-0.20mm unless called out.

CAD is nominal. Fit, measure and qualify before calibration.

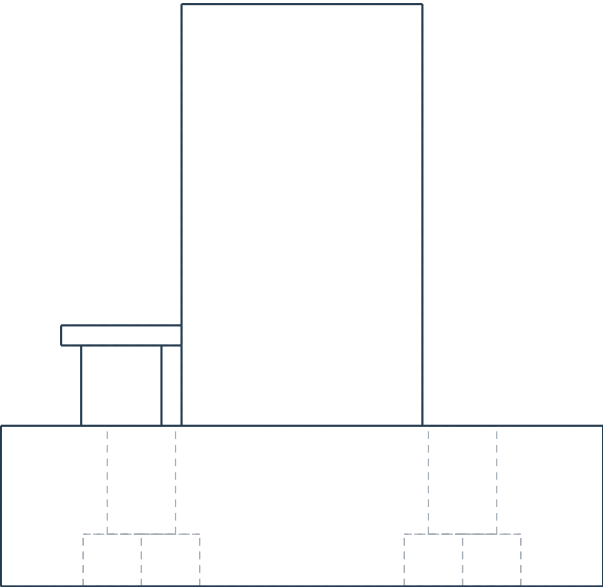
P02 | P02 BLOCK1 CARRIAGE

Prototype Rev A | Units mm | Dimensions control; do not scale printed pages



TOP (view scale 1.67:1)

102.0000 X



END (view scale 2.66:1)



FRONT (view scale 1.67:1)

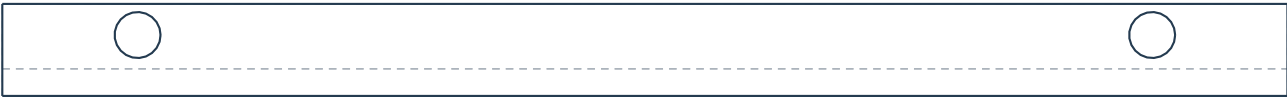
- BODY 102 x30 x8; print flat with push tower up.
- Cassette holes at X8 Y+/-8: DIA3.40 through.
- Underside nut recess AF5.80 x2.60 at both holes.
- Push face X102; micrometer contacts local Z24, Y0.
- Slide width30.00 in30.60 channel; remove burrs only.
- Assembled bottom Z6; top Z14. Check free motion.

General prototype target: +/-0.20mm unless called out.

CAD is nominal. Fit, measure and qualify before calibration.

P03 | P03 KEEPER QTY2

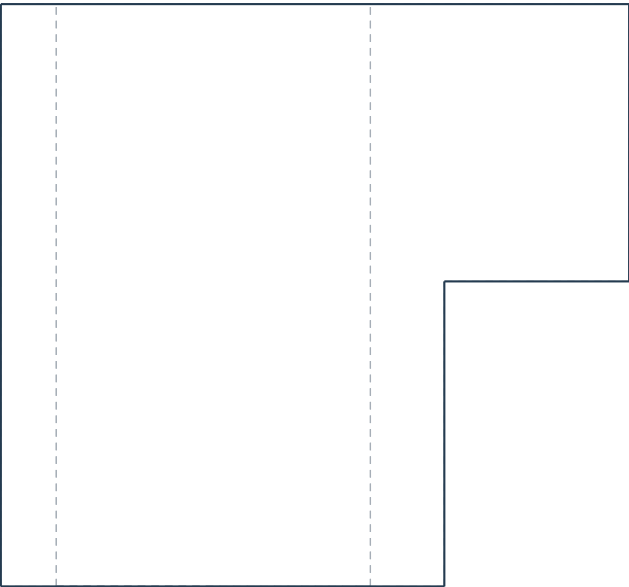
Prototype Rev A | Units mm | Dimensions control; do not scale printed pages



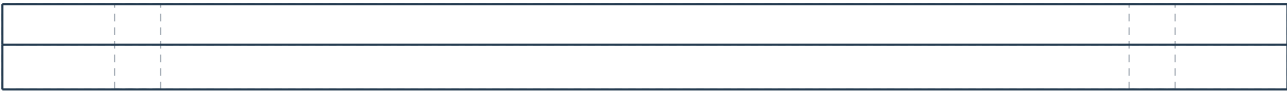
6.80 Y

TOP (view scale 1.79:1)

95.0000 X



END (view scale 12.22:1)



6.30 Z

FRONT (view scale 1.79:1)

PRINT 2 IDENTICAL; second rotates180deg about Z.

Length95; width6.80; height6.30.

Overhang2.00; underside clearance step3.30.

DIA3.40 holes at X10 and85, Y2.50.

Assemble one at (45,15.50,11); mate at (140,-15.50,11).

Nominal top clearance to slide0.30; shim/adjust after print.

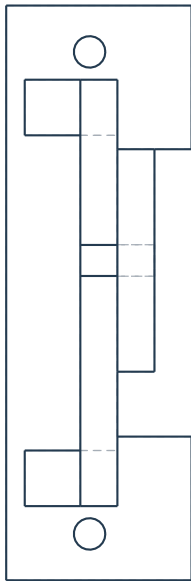
General prototype target: +/-0.20mm unless called out.

CAD is nominal. Fit, measure and qualify before calibration.

P04 | P04 BLOCK2 SENSOR MOUNT

Prototype Rev A | Units mm | Dimensions control; do not scale printed pages

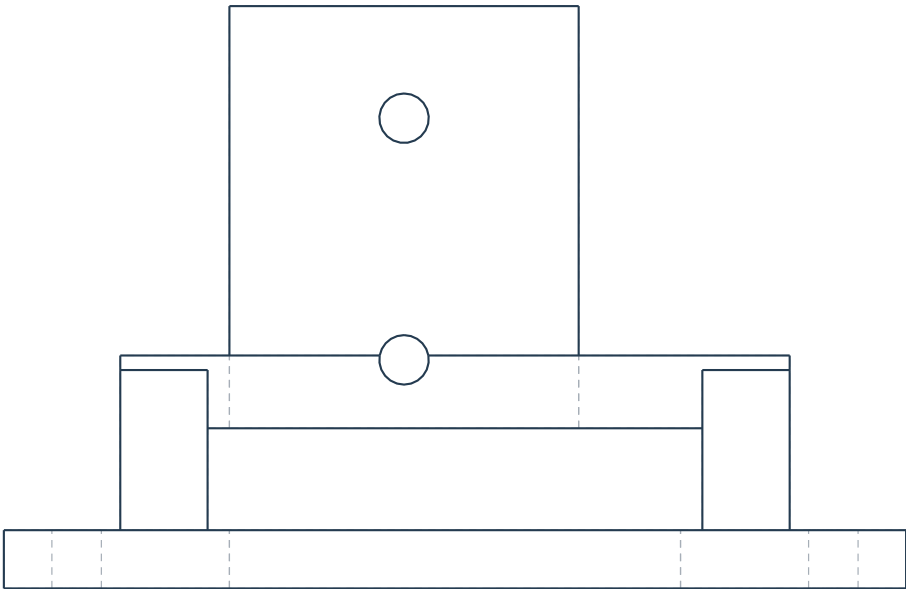
TOP (view scale 1.23:1)



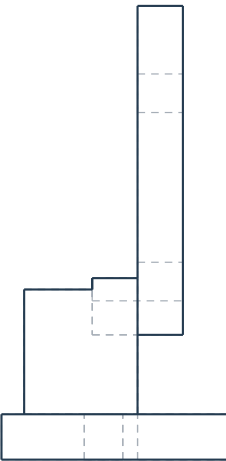
20.0000 X

62.00 Y

END (view scale 1.93:1)



FRONT (view scale 1.50:1)



40.00 Z

Flange20 x62 x4 at X13 Y-31 Z6; print base flat.

Sensor back seating face X29; support plate4 thick.

DIA3.40 sensor holes: Y3.50; Z21.70 and38.30.

Sensor nominal3 thick; illustrated active housing face X32.

Center sensitivityY0,Z30; housingcenterY3.50,Z30.

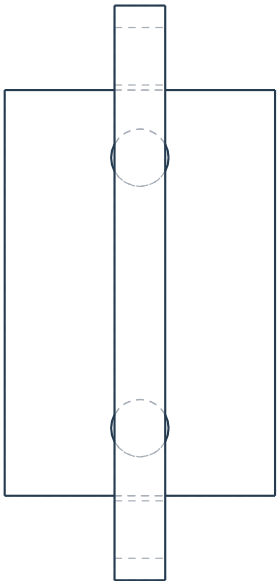
Cable exits+Y. Verify correct active face/polarity on bench.

General prototype target: +/-0.20mm unless called out.

CAD is nominal. Fit, measure and qualify before calibration.

P05 | P05 SPACER 3MM

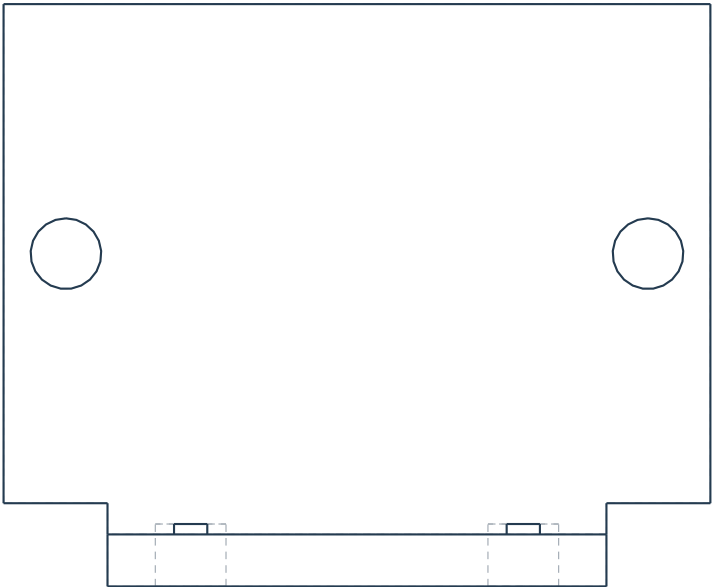
Prototype Rev A | Units mm | Dimensions control; do not scale printed pages



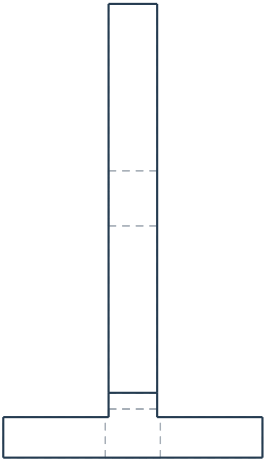
TOP (view scale 2.24:1)

16.0000 X

34.00 Y



END (view scale 2.75:1)



FRONT (view scale 2.14:1)

28.00 Z

SPACER 3.00 thick; controls nominal inner-face separation.

Plate 34 wide; Z18 to 42; magnet axis Y0, Z30.

Foot 16 x 24 x 2.50; bottom Z14; neck 24 wide to Z18.

Foot DIA 3.40 holes at X0, Y +/- 8. Use M3x10 screws.

Cup clamp DIA 3.40 holes along X at Y +/- 14, Z30.

Printed proxy for static tests; measure actual seated spacing.

General prototype target: +/- 0.20mm unless called out.

CAD is nominal. Fit, measure and qualify before calibration.

P06 | P06 MAGNET CUP QTY2

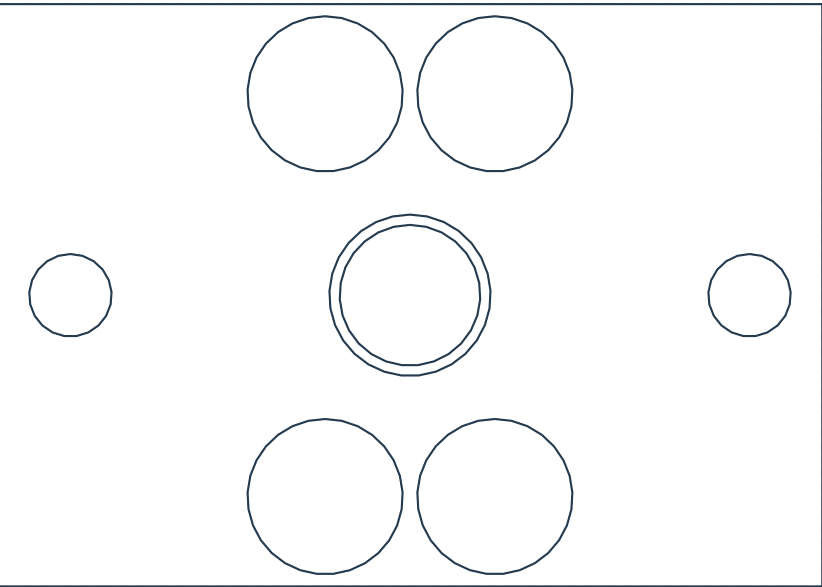
Prototype Rev A | Units mm | Dimensions control; do not scale printed pages

TOP (view scale 2.24:1)



1.9882 X

34.00 Y



END (view scale 3.21:1)

FRONT (view scale 2.50:1)



24.00 Z

PRINT 2; outer retaining face flat, pocket upward.

Envelope34 x24 x1.9875; magnet DIA6.35 x1.5875.

Pocket DIA6.65, depth1.5875; aperture DIA5.80, lip0.40.

2x DIA3.40 clamp holes at Y+/-14,Z0.

4x DIA6.40 clearance holes atY+/-3.50,Z+/-8.30.

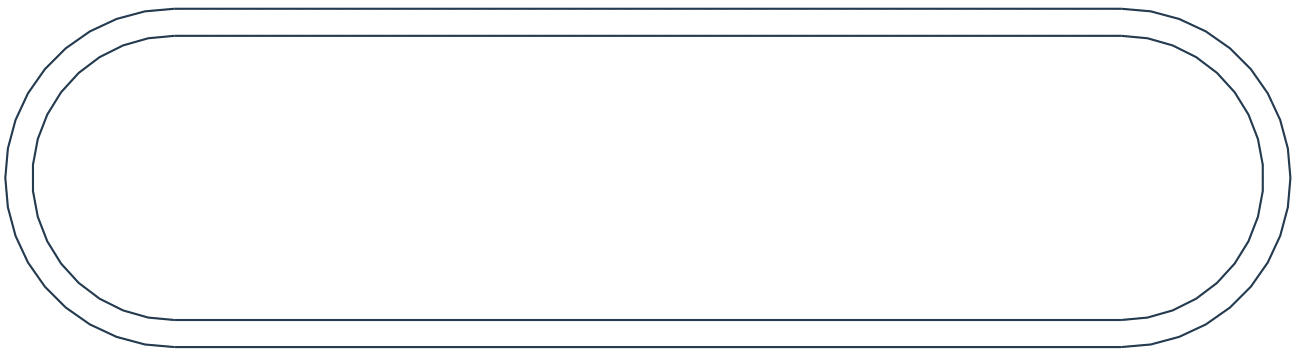
Fit gently; adjust pocket depth to measured magnet thickness.

General prototype target: +/-0.20mm unless called out.

CAD is nominal. Fit, measure and qualify before calibration.

P07 | P07 TPU RETURN BAND

Prototype Rev A | Units mm | Dimensions control; do not scale printed pages



TOP (view scale 4.47:1)

38.0000 X

10.00 Y



END (view scale 17.00:1)



FRONT (view scale 4.47:1)

0.80 Z

OPTION: TPU95A; flat print,0.80 total thickness.

Outer capsule38 x10; inner36.40 x8.40.

0.80 nominal wall. Ends stretch over printed rear posts.

Maintain light contact with micrometer, avoid high preload.

Neutral model supplied; stretched shape is illustrative.

If return sticks, use light manual preload and quantify repeatability.

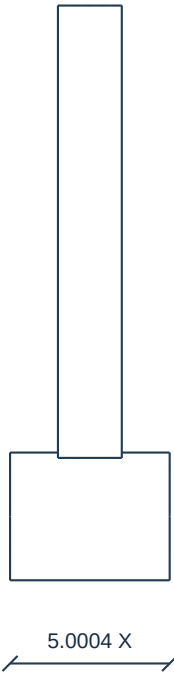
General prototype target: +/-0.20mm unless called out.

CAD is nominal. Fit, measure and qualify before calibration.

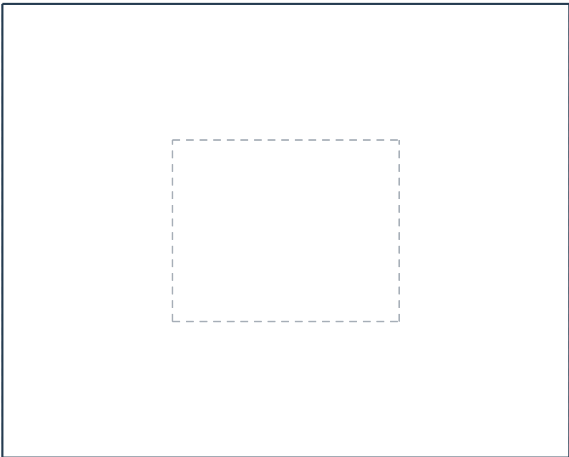
P08 | P08 5MM REFERENCE GAUGE

Prototype Rev A | Units mm | Dimensions control; do not scale printed pages

TOP (view scale 4.22:1)



FRONT (view scale 15.00:1)



18.00 Y



END (view scale 9.45:1)

4.00 Z

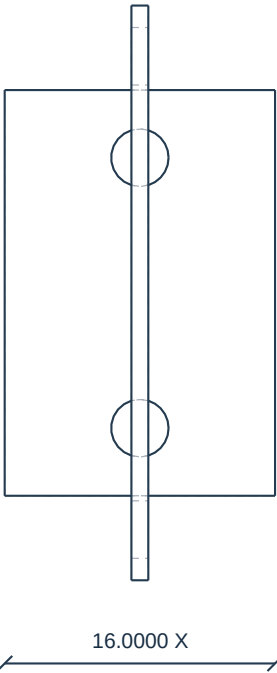
- NOMINAL contact separation5.00 alongX; tip DIA4.00.
- Handle2.00 thick,16 reach; print on broad handle face.
- Measure actual tip separation with calibrated caliper.
- Fits inside5.80 cup aperture to contact magnet face.
- Remove gauge and metal caliper before switching trials.
- Printed nominal dimension is NOT a calibrated standard.

General prototype target: +/-0.20mm unless called out.

CAD is nominal. Fit, measure and qualify before calibration.

P05A | P05A SPACER 1MM OPTION

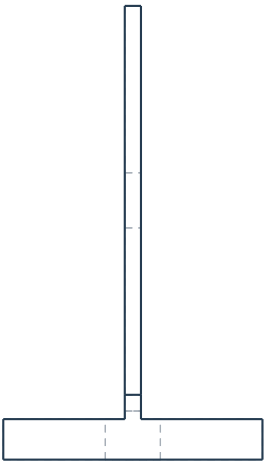
Prototype Rev A | Units mm | Dimensions control; do not scale printed pages



TOP (view scale 2.24:1)



END (view scale 2.75:1)



FRONT (view scale 2.14:1)

OPTIONAL spacer1.00 thick; all other dims P05.

Regenerate assembly if selected; measure actual seating.

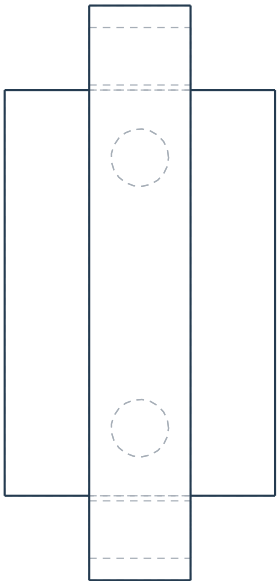
Use with both south poles outward.

General prototype target: +/-0.20mm unless called out.

CAD is nominal. Fit, measure and qualify before calibration.

P05B | P05B SPACER 6MM OPTION

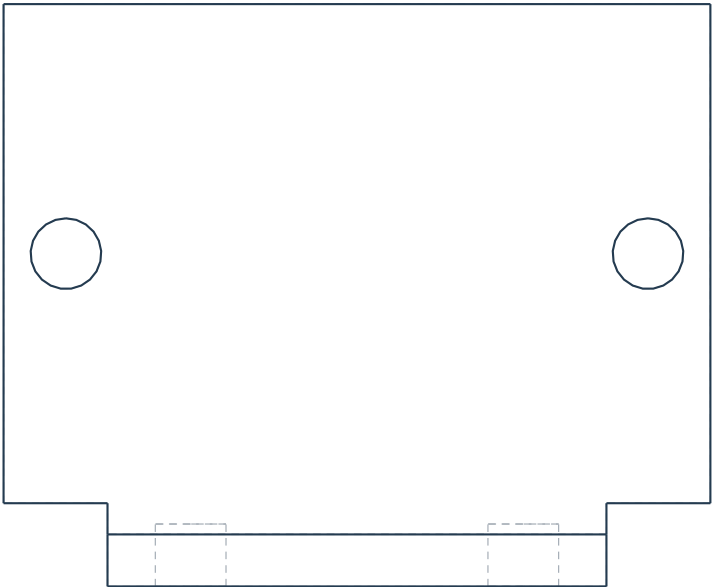
Prototype Rev A | Units mm | Dimensions control; do not scale printed pages



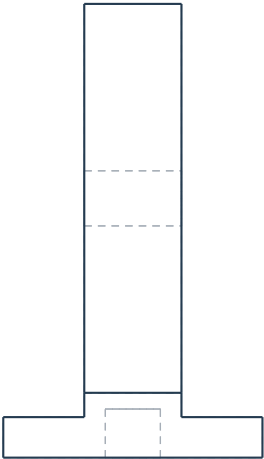
TOP (view scale 2.24:1)

16.0000 X

34.00 Y



END (view scale 2.75:1)



FRONT (view scale 2.14:1)

28.00 Z

OPTIONAL spacer6.00 thick; all other dims P05.

Use M3x20 cup screws when stack requires longer thread.

Regenerate assembly if selected; measure actual seating.

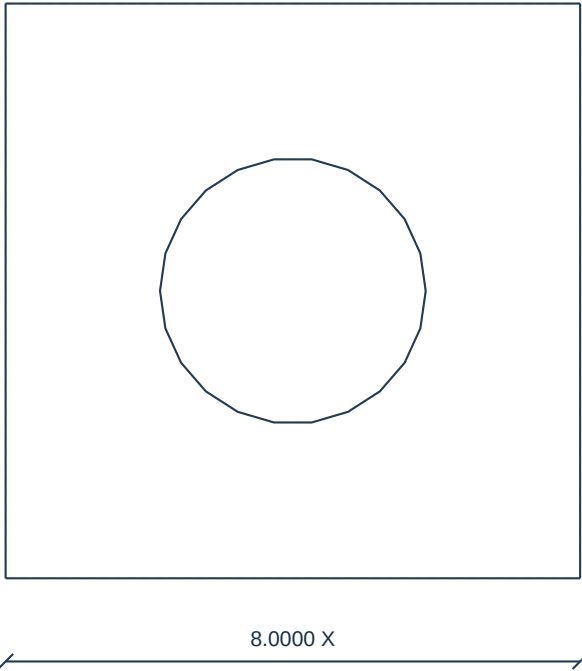
General prototype target: +/-0.20mm unless called out.

CAD is nominal. Fit, measure and qualify before calibration.

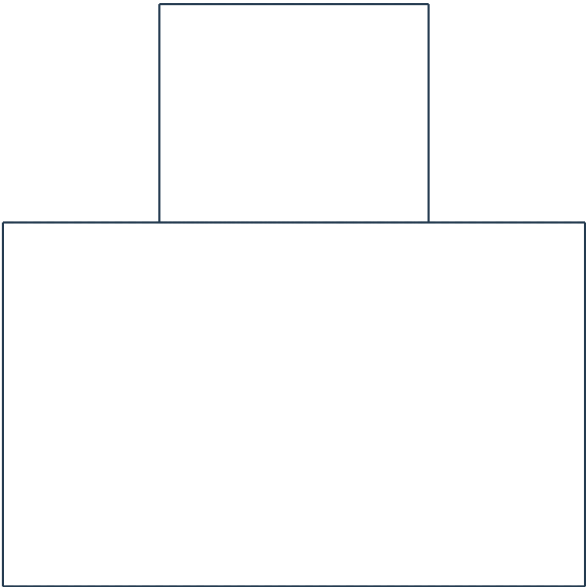
P09 | P09 TABLE FOOT QTY4

Prototype Rev A | Units mm | Dimensions control; do not scale printed pages

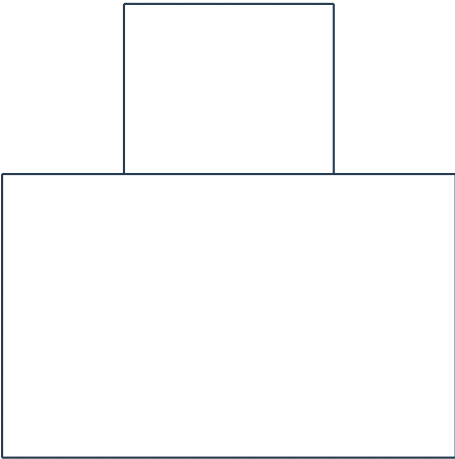
TOP (view scale 9.50:1)



END (view scale 9.62:1)



FRONT (view scale 7.50:1)



8.00 Z

- PRINT4; pad8 x8 x5, locating pin DIA3.70 x3.
- Seat underneath base at X8,180; Y+/-32.
- Target light push fit; adjust pin to your printer.
- Feet provide clearance for protruding nylon screw ends.
- Assembly bottom Z-5; base underside remains Z0.
- Do not omit feet unless shortening screws appropriately.

General prototype target: +/-0.20mm unless called out.

CAD is nominal. Fit, measure and qualify before calibration.

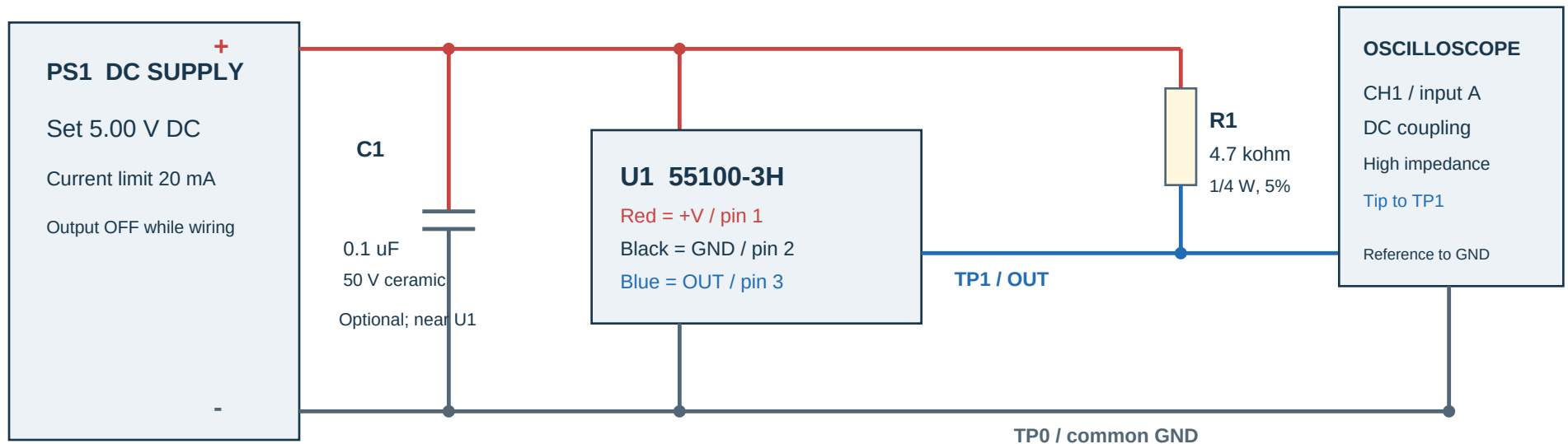
Sensor circuit and oscilloscope

5.00 V bench circuit | one sensor per channel | magnetic ON is a LOW voltage

MAGNETIC AND ELECTRICAL STATES

South pole presented to active face; B exceeds operate threshold: ON -> OUT LOW.

Withdraw until B falls below release threshold: OFF -> OUT pulled HIGH toward +5 V.



Pullup check at 5 V: $I \leq 5 / 4700 = 1.064 \text{ mA}$; $P \leq 25 / 4700 = 0.00532 \text{ W}$.

At +5% supply and -5% resistor: $I \leq 1.176 \text{ mA}$, $P \leq 0.00618 \text{ W}$. A 1/4 W resistor is ample.

Use $\geq 1 \text{ Mohm}$ scope input; 1x supplied Fluke leads or matched 10x probe. Do not use 50 ohm termination.

Start at 1 V/div, 100 ms/div, falling trigger at 2.5 V for approach; rising trigger for withdrawal.

Connect every probe ground to TP0 only. For two sensors duplicate U1/R1/C1; keep the outputs separate.

Manufacturer basis: Littelfuse 55100 datasheet, GD.05/16/23, pp. 2-3. Connections are red/black/blue.

Breadboard placement and node map

McMaster 1388N14, 830 contact numbered breadboard with jumpers | rows 1-22 shown | power rails unused

PS1 +5 V -> a5

U1 SENSOR HARNESS

Red lead -> b5

Black lead -> b15

Blue lead -> b10

U2 OPTIONAL HARNESS

Red -> c25; jumper d5 to a25

Black -> b27; jumper d15 to a27

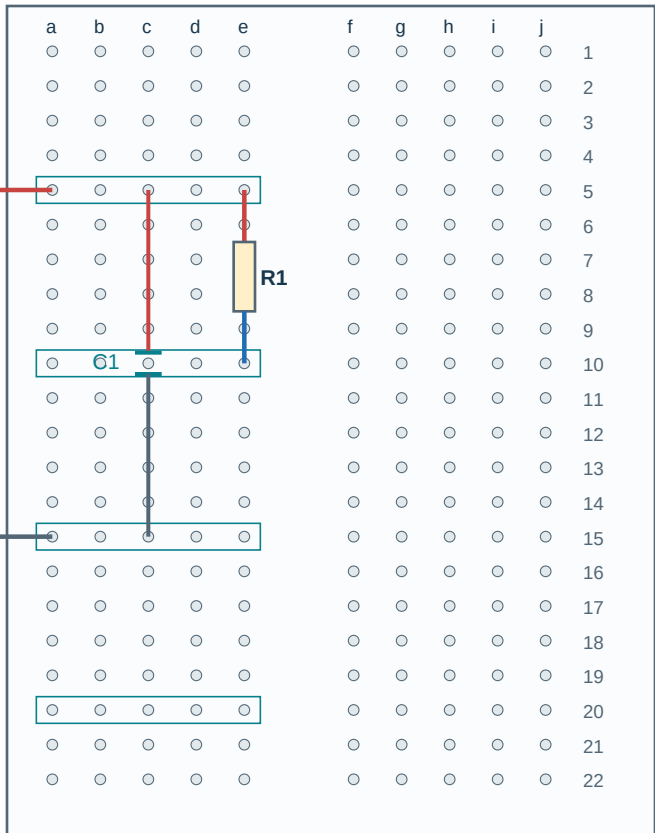
Blue -> b20

Optional rows 25/27 below crop.

C2: d25 to d27 (optional).

CH2 reference -> c27.

PS1 GND -> a15



Center trench: e and f are isolated.

NODE SCHEDULE

a5-e5 +5 V

a10-e10 U1 / CH1 OUT

a15-e15 GND

a20-e20 U2 / CH2 OUT

CH1 tip -> a10

CH1 reference -> e15

R1: e5 to e10

C1: c5 to c15

Optional channel 2:

R2: b25 to e20; tip -> a20

Each a-e group on a single numbered row is connected internally. Adjacent rows are isolated.

Use one U1 first. R2 and all U2 connections are omitted for single-sensor calibration.

Sensor body stays in block2. Lead ends go to breadboard; strain-relieve the cable away from the carriage.

Power OFF: check row continuity and supply-to-ground resistance. Power ON: verify HIGH before approach.

R1/R2 = 4.7 kohm. C1 = 0.1 uF. A breadboard capacitor at the cable end may not suppress sensor-end spikes.

Calibrate the sensor.

Then qualify the assembly.

A practical bench procedure for the two south-out disk magnets and Littelfuse 55100-3H. Use the accompanying E-01/E-02 drawings and mechanical fixture drawing set.

1. Define the states before testing

This procedure calls a magnetically actuated sensor **ON**, which appears as a LOW output voltage. The confirmed state table is max ON / min OFF at maximum, both OFF in the intermediate region, and max OFF / min ON at minimum. Test the complete stroke in both directions because operate and release occur at different field strengths.

Location	Magnetic state: max / min	Electrical output: CH1 / CH2
Horizontal + maximum	ON / OFF	LOW / HIGH
Intermediate	OFF / OFF	HIGH / HIGH
Horizontal - minimum	OFF / ON	HIGH / LOW

2. Assemble and prove the circuit

Use 55100-3H-02-A tinned leads, or verify the exact termination on your existing sensor. With output disabled, set the supply to 5.00 V and limit current to 20 mA for one sensor or 30 mA for two. Follow E-01 and E-02; do not join the blue outputs. Connect scope references only to circuit ground. The supplied 1x Fluke test leads are suitable for this slow test; configure attenuation to match any substituted probe.

Enable the supply with the magnet far away. Verify approximately 5 V at OUT. Approach the known south face toward the marked active sensor face. Verify a clear transition toward 0 V. If absent, check wiring, ground continuity, pullup, pole direction, and alignment. Reverse direction to prove release. Check your own HIGH and LOW levels; the reference maximum LOW is 0.4 V at 20 mA [1].

The 3H supply range is 2.7-24 V, operating temperature -40 to +100 C, and supply current 1.1-2.4 mA. The 2023 datasheet lists 55 G sensitivity and approximately 19 mm actuation with H-58, a 21 x 7 x 4.7 mm actuator [1]. Use measured thresholds for the small disk magnets; 19 mm is not their specified working distance.

Scope startup settings

DC coupling, high impedance input (at least 1 Mohm), approximately 1 V/div and 100 ms/div. Trigger at 2.5 V: falling edge for operate, rising edge for release. The voltage trigger marks the digital edge; it is not the magnetic threshold. Save both channel traces for the final stroke test.

Measure distance and sensitivity

Use one reference magnet first. Then retain both assembly magnets for pair verification.

3. Establish the geometry and reference

Block1 carries the pair with north faces inward and south faces outward. Set the spacer to the measured production separation between the inward magnet faces, including adhesive layers. The supplied CAD default is 3.00 mm. A printed spacer is a static magnetic surrogate for aluminum; it does not reproduce production heat transfer or dynamic eddy-current effects.

Mount the sensor in block2 with its active broad face toward the tested south face. The sensitivity center is at the 25.5 mm body longitudinal midpoint and nominally 2.00 mm from the side opposite cable exit; confirm by a small lateral sweep. The 2.00 mm dimension is lateral, not sensing depth. Record sensor orientation. The axial distance is **$z = \text{measured face gap } g + \text{Hall sensing-depth offset } a$** . Use the same definition in calibration and assembly calculations; do not substitute carriage travel or overall case thickness for g or a . Obtain a from the supplier or a multi-distance field fit.

Use the micrometer only as an incremental actuator. The specified 0-13 mm travel and 0.01 mm graduation do not establish fixture accuracy. Establish a non-contact zero or a known nonmagnetic gap standard; check actual displacement with a calibrated instrument and include zero, alignment, backlash and repeatability in the uncertainty. Keep ferrous instruments away while reading switching state. Keep positive preload on the drive and approach each bracket from a consistent direction.

4. Measure each sensor with the standard magnet

Measure the reference disk diameter and thickness. At the reference temperature, measure the center-face gauss before assembly with a calibrated probe, known lift-off, fixed orientation and unchanged positioning. A contact reading is not a zero-depth reading: record the Hall element depth inside the probe. The calculation sheet must use reference dimensions and that lift-off. Do not mix an H-58 rectangular magnet with the disk equation.

Start with the sensor OFF at a large gap. Decrease g in 0.25 mm steps until it turns ON; return outside the transition and bracket in 0.01 mm steps, or coarser if the verified uncertainty requires it. Record the last OFF gap and first ON gap. Continue slightly closer, then increase g and record the last ON / first OFF release bracket. Repeat at least five complete cycles per sensor. Never infer release from the operate measurement.

5. Record the conservative envelopes

For guaranteed operation, use the smallest observed approach distance after subtracting the distance uncertainty. For guaranteed release, use the largest observed withdrawal distance after adding uncertainty. Enter individual measured distances in the workbook as its labels specify. Use the **largest operate field** and **smallest release field** across repeats, supply conditions and temperature to check guaranteed ON and OFF. A threshold inferred from a field model is an estimate until checked with a field probe at the sensor element location.

Verify the pair and actual travel

Keep the two magnets identified as M1 and M2 throughout calibration, assembly and aging checks.

6. Measure the actual two-magnet assembly

Measure each loose magnet surface field before mounting, with the same calibrated probe method. Install both magnets and the production-equivalent spacer in block1. Measure approach and release for each sensor from each outer south face; rotate the cassette as described in the assembly drawing. Maintain the same physical sensor face toward the tested magnet. Repeat at least five cycles and compare these measured pair distances with the calculation sheet. A disagreement larger than the measurement uncertainty calls for an alignment or field-model correction before accepting a tolerance.

The opposite magnet contributes to the field at the tested face; its effect is not captured by adding the two face-gauss readings. The accompanying workbook uses a coaxial opposed-disk approximation. A lateral slide past sensors, non-coaxial centers, neighboring steel, and a moving conductive plate need measured field-versus-position data or a 3D magnetic analysis. A single face-gauss number cannot determine all of those effects.

7. Test temperature and full-stroke behavior

Repeat at the required minimum, nominal and maximum operating temperatures, measuring actual magnet and sensor temperature after stabilization. Keep electronics outside the heated region. The printed fixture material, adhesive and breadboard can limit test temperature below the sensor rating. Perform high-temperature work only with a qualified fixture material and independent distance measurement; printed-plastic expansion and creep can imitate magnetic drift.

Install the sensors in the real assembly. Observe CH1=max and CH2=min over the entire stroke in both directions, through power cycles and at operating speed. Verify the selected truth table, endpoint ON margins, OFF margins, and the width of the both-OFF region. Test worst allowed assembly offsets and supply variation. The axial fixture calibrates threshold; it does not reproduce the unspecified production stroke geometry.

8. Obtain a defensible life input

Retain the assembled pair in the actual operating thermal and magnetic environment for known elapsed times. At each interval, return to the same reference temperature and remeasure face field and switching distances using the same datums. Include repeated measurements and sample-to-sample variation. Retain a protected reference sample to detect instrument drift. Record irreversible loss separately from reversible temperature loss and from spacer, adhesive or printed-part movement.

Use the workbook life result only after supplying a measured aging law or rate and its time units. Extrapolation beyond the test duration is conditional; validate the model and use an upper loss-rate bound. Grade, surface gauss and temperature alone do not determine elapsed time to failure. The endpoint fails when the available field at its worst working position falls below the required operate threshold, even if an already-ON sensor has not yet released.

Test record supplied as CSV

The blank **Calibration_Test_Log.csv** has fields for sensor/magnet IDs, temperature, supply, reference geometry and gauss, face-gap brackets, uncertainty, orientation and elapsed aging time. It is a bench record; transfer verified summary inputs to the calculation workbook. No measured data have been invented.

Electrical purchase list and sources

All added electrical bench items below are COTS offered by McMaster. Sensor and magnets are the specified articles under test. Mechanical items are listed in the fixture package.

Ref. / qty	Item	McMaster item / source
PS1 / 1	Adjustable DC bench supply, 1-36 V, 0-3 A; set to 5 V	5772N11 [2]
SC1 / 1	Fluke 123B, two-channel oscilloscope/multimeter; leads and clips included	6587K41 [3]
BB1 / 1	830-contact numbered no-solder board; 70 jumper wires included	1388N14 [4]
R1, R2 / 2	4.7 kohm, 0.25 W, 5% carbon-film pullups; buy required package	1348N227 [5]
C1, C2 / 2 optional	0.1 uF, 50 V ceramic; package quantity 5	4861N11 [6]
Leads / 2	Sleeved alligator clip to stackable standard banana plug, 36 in; one red and one black	6927K42; choose length/color [7]
U1, U2 / 2 existing	Littelfuse 55100-3H sensor; tinned-lead 55100-3H-02-A used in drawing	User specified; exact sensor is not claimed McMaster stock
M1, M2 / 2 existing	Radial D1/4 x 1/16 axial N35 disk magnets	User specified; magnet is not claimed McMaster stock

Catalog numbers are checked against public catalog listings. Order 6927K42 in 36 in length, one red and one black. Clip the supply leads to individual solid jumper leads at a5 and a15; support and insulate each clip so it cannot bridge nearby nodes. Confirm stock and connector compatibility when ordering.

Sources (accessed 2026-09-09)

[1] Littelfuse 55100 datasheet, revision GD.05/16/23

[2] McMaster adjustable bench power supplies

[3] McMaster Fluke oscilloscope/multimeters

[4] McMaster numbered solderless breadboards

[5] McMaster carbon-film resistors, 4.7 kohm row

[6] McMaster circuit-board capacitors, 0.1 uF row

[7] McMaster banana-to-alligator patch cords, style B